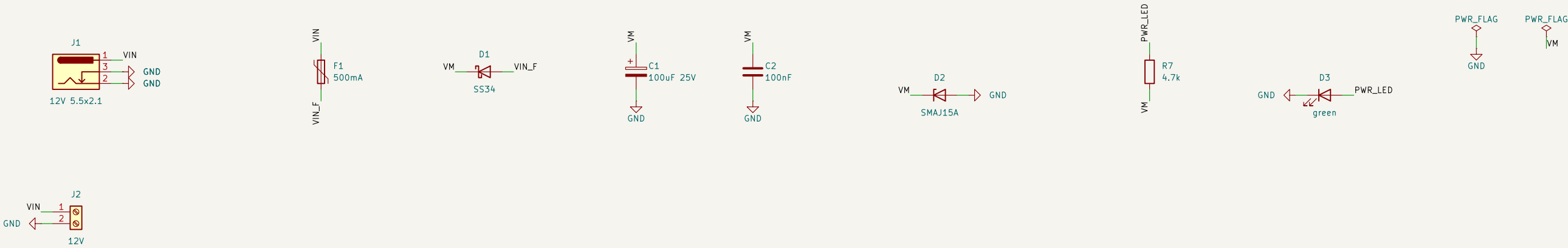


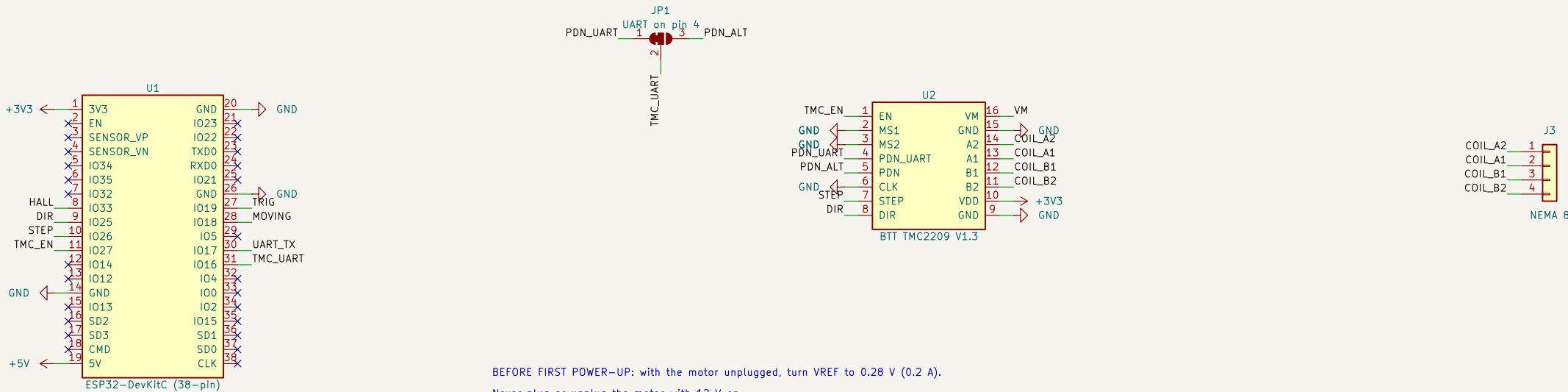
Scan-mirror controller: ESP32-DevKitC + BTT TMC2209 for the NEMA 8 mirror stepper

12 V in on J1 or J2 (ALITOVE 12 V 3 A). F1 limits it, D1 blocks reverse polarity, D2 clamps spikes under the TMC2209's 29 V limit. D3 is lit while the motor has power.

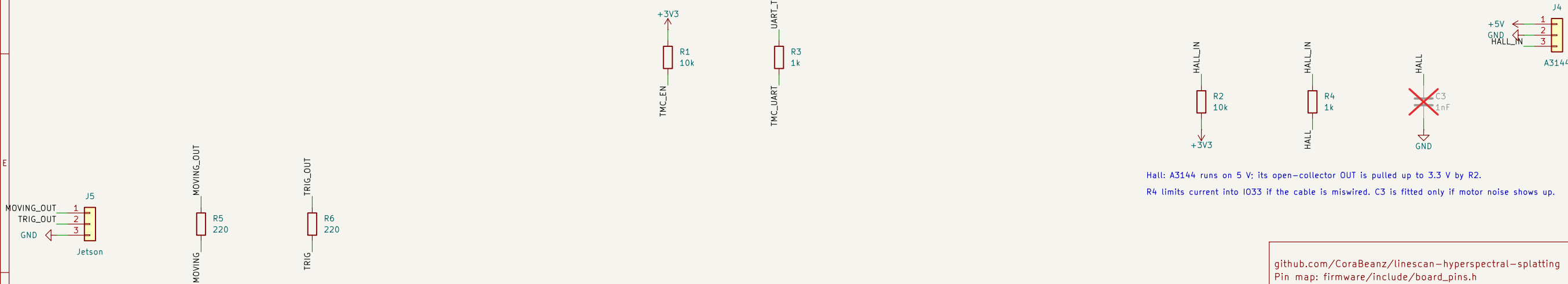


The ESP32 is powered over USB from the Jetson, which is also its command link, so there is no 5 V regulator here. With USB unplugged the TMC2209 has no logic supply (VDD), so the motor cannot move.

R1 holds EN high (motor off) while the ESP32 boots.
UART: RX (IO16) straight to PDN_UART, TX (IO17) through R3.
JP1 picks the module pin: 1-2 = pin 4 (BTT default), 2-3 = pin 5.
MS1 = MS2 = GND: UART address 0. CLK = GND: internal clock.



BEFORE FIRST POWER-UP: with the motor unplugged, turn VREF to 0.28 V (0.2 A).
Never plug or unplug the motor with 12 V on.



J5 to the Jetson's 40-pin header (3.3 V GPIO): MOVING, TRIG, GND. R5 and R6 protect both ends from a wrong pin.

Hall: A3144 runs on 5 V; its open-collector OUT is pulled up to 3.3 V by R2.
R4 limits current into IO33 if the cable is miswired. C3 is fitted only if motor noise shows up.